

ODONT 541.1

CARIOLOGY

Fluoride

Lecture #11

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Office Hours: By appointment
(Tuesday and Wednesday afternoons)

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Active Ingredients

- Mechanism of action
 - Can be dose dependent
- Formulation
 - Is it pure or mixed with another cooperative ingredient
- Vehicle of delivery
 - Affects:
 - What dose reaches the intended site vs. Elsewhere
 - What total dose is administered to achieve a therapeutic dose at the intended site
 - How long it persists at that site
 - Application frequency, ease of use, and compliance
- Indications
 - FDA approves vs. Off label
- Contraindications
 - Interactions
- Instructions For Use

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Pre-Eruptive Fluoride

- Pre eruptive fluoride (from in utero to eruption) is NOT preferred
- During amelogenesis, fluoride can be incorporated into enamel crystals
- In correct doses, it can be minimally beneficial without fluorosis.
 - However, we find more therapeutic effect in post eruptive topical fluoride
 - Even with optimal pre eruptive fluoridation, post eruptive fluoridation is still necessary to maintain anti caries benefit.
 - Therefore we do not Rx or supplement with the goal of pre eruptive fluoride incorporation
- Fluorosis
 - Excess dosage of fluoride during development disrupts normal ameloblast function, leading to hypomineralization
 - Mild: White striations
 - Moderate: Opacities
 - Severe: Pitting
 - Uncommon in the US. Suggestive of birth/childhood elsewhere.
 - Generally considered a cosmetic concern. Though surface enamel may be more porous in severe cases, it is still more resistant to demineralization

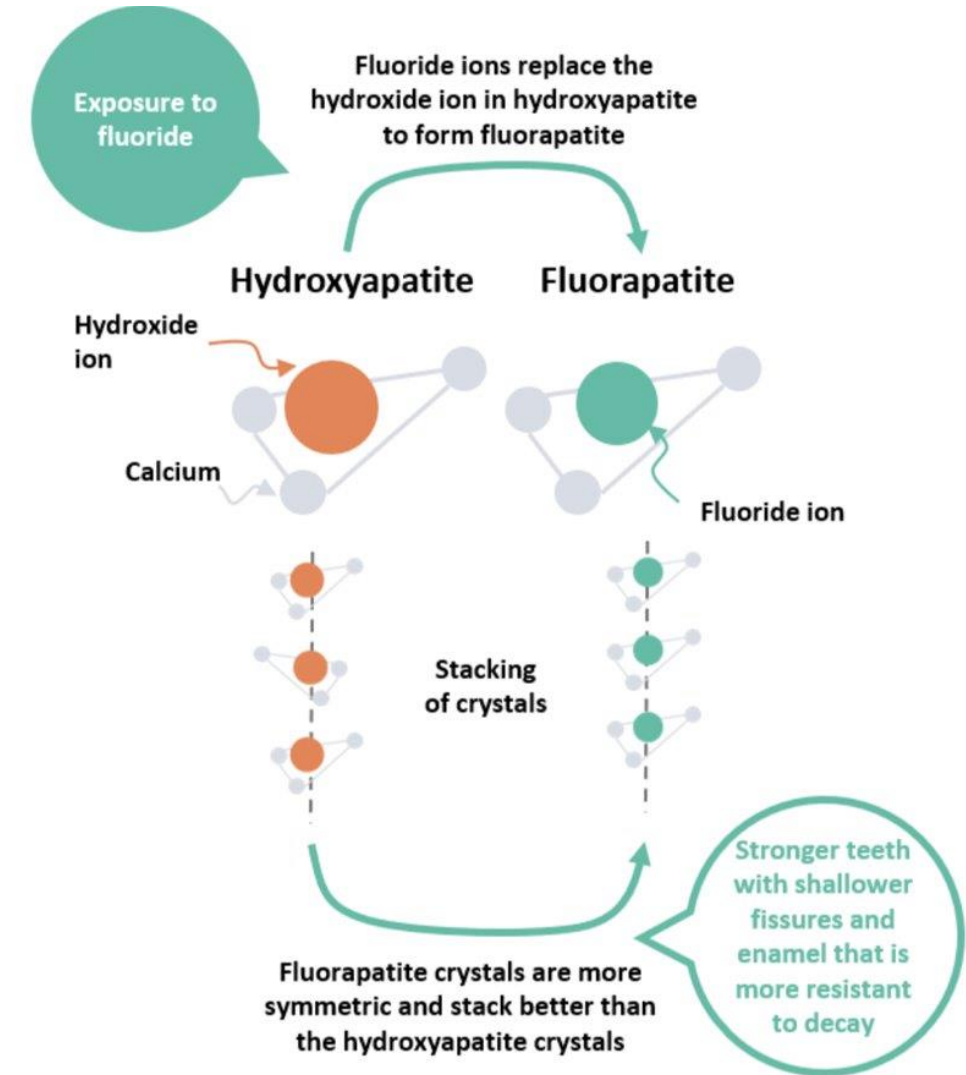
Mechanism 1: Inhibit Demineralization

- Fluoride in plaque fluid adsorbs to crystal surface
- Created fluoridated hydroxyapatite
- Changes critical pH from about 5.5 to about 4.5, making enamel resistant to demineralization on acid exposure
 - o Remember that critical pH is not the exact same between patients. It is partly a reflection of surface enamel composition and partly a reflection of mineral saturation in saliva.



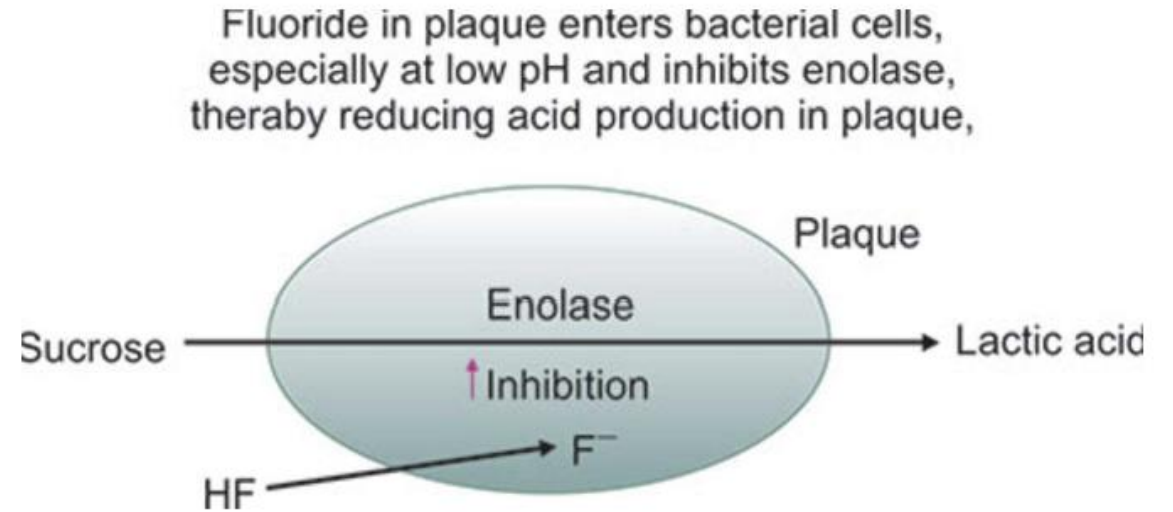
Mechanism 2: Enhances Remineralization

- Fluoride attracts calcium and phosphate faster (because more electronegative)
 - This promotes faster crystal regrowth
 - And forms larger, more acid resistant crystals
- High concentrations of fluoride (varnishes and Rx toothpastes) create **globules of calcium fluoride (CaF_2) on the enamel**
 - **This acts as a fluoride reservoir, releasing fluoride during acid attacks.**
 - Remember that there is a maximum uptake rate of fluoride, which is why we generally recommend frequent low dose exposure (OTC toothpaste, fluoridated water, etc)
 - However, F- is most effective within a specific pH range, which may not be present at during the above F- exposures.
 - High caries risk patients often pH cycle with frequent snacking/drinking.
 - Fluoride needs to be localized to the enamel and available when that pH cycling occurs for optimal effect.
 - Formation of fluoride reservoir is a beneficial "smart release" mechanism.



Mechanism 3: Antimicrobial Effects

- Inhibits bacterial enzymes
 - Enolase inhibition
 - Enolase is the enzyme that catalyzes the ninth/last step of glycolysis
 - Fluoride + Magnesium + Phosphate form a competitive inhibitor that blocks enolase active site
 - Reducing glycolysis -> less acid production
- This effect is modest at OTC concentrations but more pronounced with varnish/Rx concentrations.



Mechanism 4: Weak Acid Effect

- **F- ions alone can not cross the bacterial wall**
- But when bacteria produce acid (H^+), F^- becomes HF which CAN cross the bacterial wall
- **Once inside the bacterium, the HF dissociates into H^+ and F^- again**
 - H^+ acidifies the interior of the cell
 - F^- compromises the function of F-ATPases, which normally function to export H^+ out of the cell
 - So H^+ stays stuck inside the cell because F^- made it into the cell
 - Too much H^+ in the cell $\rightarrow$ cytoplasmic acidification $\rightarrow$ acid inhibition of glycolytic enzymes
- **This is why fluorides ANTICARIOGENIC effect is most effective at slightly acidic pH levels**
 - Fluorides REMIN effect is also most effective at slightly acidic pH levels

High vs. Low Dose

- For most patients, frequent low-dose fluoride exposure is preferred
 - Demineralization happens all day every day.
 - Primary prevention is most effective all day every day
 - Advise patients to maximize retention time. Avoid rinsing aggressively after brushing.
- For high risk patients, higher doses may be indicated. High risk patients:
 - Are often not achieving frequent low dose exposure (not brushing well)
 - May already have incipient lesions
 - Secondary prevention may be necessary
 - Pt may already have white spot lesions that need remineralization.
 - Those WSLs might be covered in plaque
 - High dose fluoride varnish is the best at creating the fluoride reservoir effect despite the presence of plaque.
 - Remember, topical fluoride can not cause fluorosis. Only systemic does and only during development.
- At most standard doses, fluoride is bacteriostatic. It lowers the activity of cariogenic bacteria
- At high dosages (SDF, fluoride varnishes), it may be temporarily bacteriocidal
 - If enough intracellular H⁺ builds up, cariogenic bacteria may die.
 - With SDF, most of the bacteriocidal effect comes from the silver, not the fluoride.



Common Fluoride Formulations

- OTC Toothpaste/rinses

- Neutral Sodium Fluoride - NaF
 - Neutral pH
 - Stable and widely used in many vehicles.
 - Standard baseline fluoride for prevention
- Stannous Fluoride – SnF₂
 - Tin has some antimicrobial and antigingivitis effects
 - But can also cause staining and have slightly metallic taste, less so in modern stabilized versions
- Sodium Monofluorophosphate Na₂PO₃F
 - A covalently bound fluoride that requires enzymatic hydrolysis to release fluoride
 - This makes it more shelf stable than NaF

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Common Fluoride Formulations

- OTC Toothpaste/rinses
 - Acidulated Phosphate Fluoride (APF)
 - F- acidulated with H_3PO_4
 - Common in high dose gel and foam formulations
 - APF is less popular now as gel/foams have been replaced with varnishes
 - Has a low pH of 3-4 that enhances enamel uptake
 - Can etch glass/ceramic and composites
 - Amine Fluoride – $R-NH-(CH_2)_n-F$
 - Organic fluoride compound more common in Europe
 - Surface active (adheres to enamel better)

Common Fluoride Formulations

- Silver Diamine Fluoride – $\text{Ag}(\text{NH}_3)_2\text{F}$
 - Silver is strong antimicrobial
 - Ammonia is a stabilizer
 - Used to arrest active caries non invasively
 - Results in permanent black staining of the carious lesion
 - Has a pH of 10-12



Public Water Fluoridation

- When first introduced in the 1940-60s, correlated with a 50-70% decrease in childhood caries. At that time:
 - Fluoridated toothpaste was not available
 - The public did not associate high sugar exposure with caries/systemic diseases.
- Now that we have widespread OTC fluoride toothpaste, diet awareness, F- varnish, preventive dentistry such as sealants etc:
 - Community water fluoridation is now associated with a 20-40% reduction in childhood caries.
- About 75% of the US public water system is fluoridated
 - 30% is naturally occurring without any added
 - The rest have added fluoride to adjust to optimal levels
- The CDC considers public water fluoridation one of the top ten public health innovations of the 20th century

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Fluoride Water Supplementation

- **Ideal concentration = 0.7ppm in public water as of 2015**
 - Ideal concentration used to be higher (1.0-1.2ppm). But then OTC fluoridated toothpaste became common
 - Warmer climates (tropics) recommend lower ideal concentrations due to higher average water consumption
- In the US, costs about \$1/person/year to optimally fluoridate public water
 - Saving \$20-40 in dental costs
- In areas where public water is unsafe or not optimally fluoridated
 - In the past, supplementation with pills or drops was recommended
 - Compliance is poor and intake is difficult to measure
 - Bottled water (single serve and commercial) often has better compliance
- Guidelines have shifted away from supplementation and towards more professional grade FL-application (varnishes, Rx toothpaste)

Mouth Rinses

- OTC solutions are generally **230ppm** sodium fluoride
- Many contain alcohol as a solvent (concerning for some patients) but is not critical for antimicrobial effect. Alcohol free variants can be equally effective.
- Rx mouthrinses offer 920ppm for use once a week.
 - Uncommon in private practice. More common in assisted living or in school programs.
- Patient must be able to expectorate, not swallow
 - Use 10mL (two teaspoons) after brushing/flossing, swish for 60 seconds, and spit
- Sufficient contact time needed.
 - F- retained in plaque and saliva, not only direct topical effect.



OTC Fluoridated Toothpaste

- Brushing with fluoride toothpaste increases the fluoride concentration in saliva 100x to 1,000x
 - This concentration returns to baseline levels within 1 to 2 hours
- Fluoride toothpaste makes up more than 95% of toothpaste sales in the U.S.
 - **OTC** concentration of **1,000 to 1,500 ppm** (regulated)
- Which ones the best?
 - Less abrasive
 - Most biocompatible for patient specific needs
 - SDS reactions
 - Taste objections
 - Sensitivity additives
 - The one you'll use regularly and well...



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OTC Toothpastes Contain:

- Abrasives (e.g., calcium carbonate, dehydrated silica, and aluminum oxides),
- Detergent or surfactant (e.g., sodium lauryl sulfate),
- Antibacterial agents (e.g., triclosan),
- Humectants (e.g., sorbitol, glycerin),
- Flavoring agents (e.g., saccharin, peppermint, spearmint)
- Fluoride
- Preservatives (e.g., sodium benzoate, methylparaben)

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Cochrane Database of Systematic Reviews | Review - Intervention

Fluoride toothpastes of different concentrations for preventing dental caries

✉ Tanya Walsh, Helen V Worthington, Anne-Marie Glenny, Valeria CC Marinho, Ana Jeroncic Authors' declarations of interest

Version published: 04 March 2019 Version history

<https://doi.org/10.1002/14651858.CD007868.pub3>

Authors' conclusions

This Cochrane Review supports the benefits of using fluoride toothpaste in preventing caries when compared to non-fluoride toothpaste. Evidence for the effects of different fluoride concentrations is more limited, but a dose-response effect was observed for D(M)FS in children and adolescents. For many comparisons of different concentrations the caries-preventive effects and our confidence in these effect estimates are uncertain and could be challenged by further research. The choice of fluoride toothpaste concentration for young children should be balanced against the risk of fluorosis.

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Rx Toothpastes

Rx toothpaste

- **5,000 ppm fluoride**
- Keep away from kids and pets, for whom this dose of fluoride may have adverse effects
- Recommended to some patients based on their caries risk status



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Sample Prescription (PreviDent 5000 Plus)

Patient Name: _____

DOB: _____

Rx: Sodium Fluoride 1.1% dental cream
(PreviDent 5000 Plus)

Sig:

Brush teeth with a thin ribbon once daily at bedtime for 2 minutes.
Expectorate. Do not rinse, eat, or drink for 30 minutes after use.

Dispense: 1 tube (100 mL or 3.4 oz)

Refills: 3

Indication: High caries risk / root caries prevention

Prescriber: _____

DEA/NPI: _____

Signature: _____

Rx Gels/Foams

- Popular in the past but now largely replaced by varnish
 - Still used in patients who may not like the consistency of varnish or have allergy concerns
- Can be Rx for at home use or in office use
 - At home use can benefit from custom trays to wear overnight or for a few hours
 - Ex: Head and neck radiation patient may benefit from custom trays and at home use of fluoride gel to manage risk of root caries



Varnish

- Most are **22,600ppm** Sodium Fluoride
- Core mechanism is the creation of CaF_2 reservoirs
- In the past, in office gels and foams were more popular. But varnish is considered highly effective now.
- Some formularies include additional ingredients such as xylitol, calciums, or phosphates to ensure sufficient building blocks for remineralization and reservoir formation.



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Varnish

- Do not require as much compliance
 - Can be painted on quickly and over plaque, although a gauze wipe or quick toothbrushing is preferred for very heavy plaque
 - Will create a fluoride reservoir within the plaque
 - Does not require a fully dry field
 - Sticks to teeth for hours
 - Often easily applied by trained non dentists (MDs, nurses, outreach volunteers)
- Is difficult to swallow
 - Can be used on infants as soon as first tooth erupts. Safe to use for children who can not spit on request
 - Very low fluorosis risk compared to other F- products that are higher ingestion risks.
- Avoid hard/crunchy foods for 4-6 hours
- Avoid hot drinks and alcohol for 30 minutes
- Avoid brushing flossing until next day. Skip brushing tonight.

Varnish

- Many varnishes use pine tree resin (colcophony) as the adhesive base
 - True fluoride allergies are extremely rare. Most patients who think they are allergic to fluoride are actually allergic to colcophony.
 - Most common presents as contact dermatitis that begins 1-2 days after administration
 - "Any allergies to adhesives, bandages, pine nuts, or dental products?"
 - For hypersensitive patients, alternatives include:
 - Colcophony free varnishes
 - Gels, rinses, toothpastes
 - SDF
- Some formularies include calciums and phosphates to ensure sufficient building blocks for remineralization and reservoir formation.

Varnish

Cons

- Strong taste and sticky sensation can be objectionable
 - Ethanol solvent that helps the varnish spread can have alcohol odor
- Avoid hard/crunchy foods for 4-6 hours
- Avoid hot drinks for XXX
 - But you can drink cold and room temp foods right away
- Avoid brushing flossing until next day. SKIP brushing TONIGHT!
- Many varnishes use pine tree resin (colcophony) as the adhesive base
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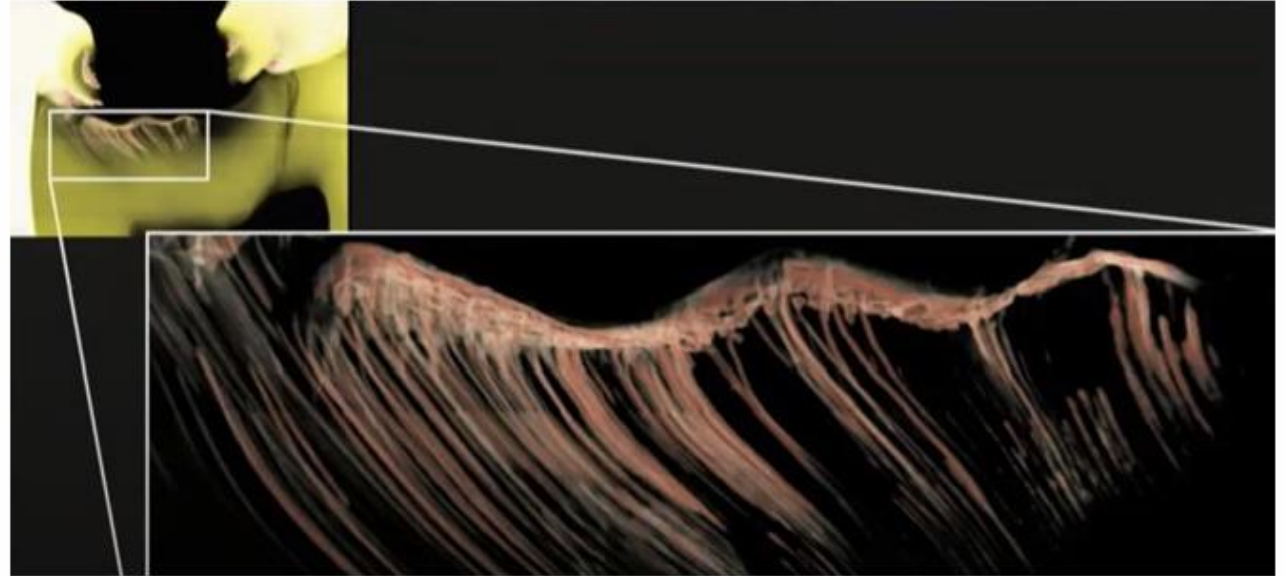
Silver Nitrate -> SDF

- 1891: Stebbings suggested that we treat caries with silver since it's antibacterial
- 1908: GV Black used silver nitrate to treat caries in children.
 - Minimally effective because it lacked fluoride (or any remineralization component). Therefore, fell out of favor.
- 1909: GV Black discovered fluoride in Colorado ground water due to local population with significant fluorosis.
- 1960s: Professor Reiichi Yamaga and his doctoral student Misuho Nishino develop SDF in Japan.
 - Japanese custom of Ohaguro may contribute to cultural acceptance, despite discoloration.
- 2014: FDA approved for dental sensitivity in the US
- 2016: Featured in New York Times, leading to boom in the US
 - **Off label use for cariostatic effect**



Silver Diamine Fluoride (SDF)

- Composed of
 - 25% silver – antimicrobial
 - 8% ammonia – keeps it in liquid solution
 - 5% fluoride – remineralization
- Indications
 - Prevent caries
 - Arrest active caries
 - **Decrease dentinal hypersensitivity**
- Mechanism of action
 - Inhibits biofilm adhesion
 - Denatures bacterial proteins
 - Breaks cell walls
 - Inhibits DNA replication
 - Occludes dentinal tubules and forms fluoroapatite
 - Also improves strength/hardness of dentin
 - Penetrates about 25 microns into enamel and 250 microns into dentin



Silver Diamine Fluoride (SDF)

- Success of SDF (and remineralization in general) requires:
 - Cleansability
 - Cavitations should either be opened or restored
 - Remineralization persistently winning over demineralization
 - Saliva can reach site
 - Reapplication of remineralizing/arresting agents like SDF every 6-12 months
- Most successful (90%) on buccal/lingual surfaces
 - Smooth surface and easily cleansable
- Medium success (80%) in pits/fissures/grooves
 - Can improve success rate by sealing/restoring over groove after applying SDF
- Least successful (70%) in interproximal areas due to more difficult cleansability by patient





SDF- Informed Consent

- Patient needs to know
 - Will PERMANENTLY stain any areas with caries OR decalcification
 - Will NOT stain healthy tooth structure
 - Will TEMPORARILY stain soft tissues
 - Regular reapplication necessary to maintain effect
 - Every 6-12 months
 - Home hygiene necessary to maintain effect
 - Surgical intervention may still be necessary

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SDF- Pediatrics

- Indications

- Low cooperation
 - May improve with time
- Preserving psyche *****
- Avoiding or deferring sedation
 - Cost
 - Backlog
 - Medical risk/candidacy/age
 - GA under 3yo more risky
 - OCS under 2yo more risky
 - Parental anxiety
- Chronologic success: Near exfoliation

BEFORE: SDF APPLIED



AFTER: ZIRCONIA CROWNS

SMART technique- Pediatrics

- Silver Modified Atraumatic Restorative Technique
 1. Minimal to no surgical preparation
 2. Treat with SDF
 3. Occlude cavitation with Glass Ionomer to aid in cleansability

CASE 3



Figure 7. A pre-op photo of occlusal decay in the mandibular second molar.



Figure 8. The application of SDF.



Figure 9. After the application of SDF.



Figure 10. The application of fluoride varnish (Profluorid Varnish [VOCO]).



Figure 11. A post-op photo of the SMART molar, SDF, and glass ionomer.

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SDF- Geriatrics

- Indications:

- Low cooperation
- Low tolerance
- Medically high risk
- Lack of homecare/ poor prognosis of restorative treatment
- Lack of transportation
- Prioritize comfort/function over cosmetics
- Cost
- Chronologic success
 - Until health improves
 - Hospice care
- Minimize aerosols for pts at high risk of respiratory complications



SDF- Everyone else

SDF Application to Interproximal Surfaces with Spongy Floss



Clinical Appearance of SDF Treated Small Interproximal Lesions



Pros

- Quick, easy, and painless
- Non-invasive
- Remineralizes your natural tooth structure
- Prevents biofilm adhesion
- Arrests 80% of lesions when applied twice yearly
- Avoid or delay more invasive restorations

Cons

- Permanently stains decay black
- 20% of lesions fail to arrest with SDF treatment alone
- Not an option for deep cavities
- Lesions between molars are the most likely to grow and may require a restoration

* Proper diet and oral hygiene, including daily flossing, are critical for long-term success

SDF Application

- Isolate
 - Protect soft tissues and unintended/cosmetic sites from staining
- Dry
 - Remove excess saliva
- Apply
 - Do NOT need to excavate caries
 - But if patient allows, you can partially excavate
 - Saturate site with SDF using microbrush or superfloss
 - Wait one minute for it to dry on its own and seep into dentinal tubules. Do not air dry or rinse.
- Cover
 - Fl varnish
 - Increases efficacy of SDF by 10%
 - Helps contain excess SDF from spreading/staining
 - GI/restoration for SMART technique
- Reassess for efficacy at next recare
 - If working/arrested caries and was not restored, reapply SDF.
 - If caries is progressing, consider more invasive treatment

Silver Diamine Fluoride (SDF)

- Contraindications

- Cosmetics
- Silver allergy
- Patients/parents not willing to or can not improve hygiene, specially in cavitated lesions
 - Unlikely to be effective. More likely to need restoration
- Pulpitis or pulpal necrosis (too late...need definitive endo/ext)
- Large caries
 - Unlikely to arrest with use of SDF alone
 - BUT can still use in CONJUNCTION with surgical treatment.
 - SDF enhances potential for successful partial caries removal and lowers risk of needing endodontic treatment



ADA Clinical Recommendations for Topical Fluorides (2013)

<https://www.ada.org/resources/ada-library/oral-health-topics/fluoride-topical-and-systemic-supplements>

- Based on the most recent literature review and consensus, the Panel recommended the following **professionally applied or prescription strength topical fluorides for caries prevention in patients at elevated risk**:
 - 2.26% fluoride varnish (professional application only)
 - OR 1.23% fluoride ACPF gel
 - OR 5,000ppm fluoride gel or paste
 - OR 900ppm fluoride mouthrinse for patients 6+ yrs old
- 2.26% F- varnish and OTC strength fluoridated toothpaste are the only applications recommended for children under 6 yrs old.

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⋮ Mandatory Reading and Videos

⋮  AAPD Fluoride Statement 2023.pdf

⋮  SDF Chairside Guide

⋮  SDF Application Videos

⋮  NZ Fluoridation update 2013-2021 (not mandatory)